CSC3041F Practice Questions

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SECTION A: Reinforcement Learning

Question 1 Multiple Choice Questions [3]

Select the **ONE** correct answer for each of the following. Write only the letter.

a) When the discount factor $\gamma = 0$, how is the optimal policy π^* characterized? [1]

A. It maximizes the long-term expected return by accounting for the full future trajectory.

B. It chooses actions based solely on the maximization of immediate rewards.

C. It requires the policy to be stochastic to ensure sufficient exploration of the state space.

D. It remains undefined for infinite-horizon problems due to the lack of convergence.

E. It necessitates the use of Monte Carlo methods to estimate the values of episodic states.

b) An agent observes a transition (s, a, r, s'). Which expression corresponds to the **TD(0) target**? [1]

A. $r + \gamma V(s')$

B. $r + \gamma \max_{a'} Q(s', a')$

C. $r + \gamma Q(s, a)$

D. $\sum_t \gamma^t r_t$

E. $V(s) + \alpha(r + \gamma V(s') - V(s))$

c) In the DQN loss function, why is the gradient not backpropagated through the target network parameters? [1]

A. To decrease the memory footprint of the computational graph by pruning the backward pass.

B. Because the target network uses non-differentiable activation functions to handle sparse rewards

C. To treat the bootstrap estimate as a stationary target, preventing the instability of a self-referential optimization objective.

D. It prevents the learning rate from decaying prematurely by stabilizing the TD-error magnitude.

E. To maintain the validity of the Experience Replay buffer's importance sampling weights.

Question 2 Short Questions [4]

In brief, answer each of the following.

a) Why does reducing ϵ too quickly harm learning? [1]

b) Give one difference between Value Iteration and Policy Iteration with respect to the Bellman equations used to calculate the state-value function V. [1]

c) Consider the Q-Learning algorith to answer the following questions:

i) Name a method that can be used to introduce exploration. [1]

ii) Give one restriction that is placed on the action space. [1]

Question 3 RL Algorithms [7]

a) Consider the following experience tuple observed by an agent in an ϵ-greedy policy setting:

The agent observes a transition from state S_1 to S_2 taking action Right with reward 3.

The agent is currently using a learning rate $\alpha = \frac{1}{2}$ and discount factor $\gamma = \frac{1}{2}$.

The current action-value estimates are:

$$Q(S_1, \text{Right}) = 2$$

For the next state S_2, the agent observes:

$$Q(S_2, \text{Up}) = 4, \quad Q(S_2, \text{Down}) = 1, \quad Q(S_2, \text{Left}) = 3, \quad Q(S_2, \text{Right}) = 2$$

During data collection, the agent followed an ϵ-greedy policy with $\epsilon = 0.2$.

i) Calculate the updated value of $Q(S_1, \text{Right})$ using the Q-Learning update rule. [3]

ii) Suppose the agent actually selected action *Left* in state S_2 during the next step. Calculate the SARSA update for $Q(S_1, \text{Right})$. [3]

iii) Explain why the Q-Learning and SARSA updates produce different values in this example. [1]

SECTION B: Control System

Question 4 Multiple Choice Questions

Select the **ONE** correct answer for each of the following. Write only the letter.

a) What is the primary advantage of data-driven control in the context of the Fourth Industrial Revolution (4IR)? [1]

A. It focuses on reducing the total number of sensors required to manage simple, linear industrial systems.

B. It enables the control of high-dimensional, non-linear systems by utilising real-time sensor data to optimise policies directly, bypassing the "glass ceiling" of simplified linear models.

C. It ensures that all control actions are strictly limited to manually derived differential equations to maintain mathematical transparency.

D. It is primarily designed to simplify Multi-Input Multi-Output (MIMO) systems by treating them as independent Single-Input Single-Output (SISO) loops.

b) Why are " Open-Loop" systems generally considered unsuitable for environments with significant external disturbances? [1]

A. They are too computationally expensive to run in real-time.

B. They lack a feedback mechanism to sense and compensate for deviations caused by the environment.

C. They are natively non-linear and therefore unpredictable.

D. They require more sensors than closed-loop systems.

Question 5 Short Questions

In brief, answer each of the following.

a) What is the difference between a SISO and MIMO system?

b) Give an advantage of RL which makes it favourable for use over a traditional PID controller?

c) Give an example of an open-loop and a closed-loop control system, and explicitly state why these systems have these characteristics.

d) Match the term in column A with the definitions in column B.

Column A: Item	**Column B: Statement / Description**
1. Proportional (P) Term	**A.** Provides a "damping" effect by reacting to the rate of change of the error, helping to prevent the system from overshooting the target.
2. Integral (I) Term	**B.** Generates a control signal based solely on the current magnitude of the error signal; it reacts to the "present".
3. Derivative (D) Term	**C.** The tendency of a system to exceed its target set-point before returning to a stable state.
	D. Eliminates steady-state error by summing the error over time, effectively addressing the "past" performance of the system.
	E. The time required for the system output to enter and stay within a specified percentage (usually 2% or 5%) of the final set-point.